4	(a)	amplitude to amplitude = 2 x_0 = 9.8 cm x_0 = 4.9 cm	A1
	(a) (ii)	$f = \dfrac{2700}{60}$ f = 45 Hz	A1
	(a) (iii)	$v = \omega\sqrt{x_0^{\,2} - x^2}$ $= 2\pi \times 45 \times \sqrt{0.049^2 - (0.049 - 0.023)^2}$ = 12 m s^{-1}	C1 A1

	(b)	$F_{max} = ma_{max} = m\omega^2 x_0$ $= 0.64 \times (2\pi \times 45)^2 \times 0.049$ $= 2500$ N	 C1 A1
5	(a)	First law of thermodynamics states that the increase in internal energy of a	